

Course Outline: Sessions in Web Applications

Title: Sessions in Web Applications **Learnpack:** + Sesiones (NO MENCIONAR PROCESOS DE AUTENTICACIÓN) **Prerequisite:** Data Flow in React and Next.js **Target Audience:** Beginner developers who have completed React/Next.js fundamentals and understand LocalStorage/SessionStorage basics **Total Lessons:** 18 **Estimated Total Length:** ~9,000 words **Format:** Mixed (39% hands-on = 7 lessons)

Course Description

Most applications need to remember something about the user — a language preference, a shopping cart, a half-filled form. Sessions are the mechanism that makes this possible, and they don't require a login to do it. This course introduces sessions as a practical tool for building smarter, more responsive frontend experiences.

Students will learn what a session is conceptually, how it differs from other storage mechanisms they've already seen, and how to manage its full lifecycle: creating it, reading from it across interactions, and cleaning it up when it's no longer needed. Every concept is grounded in realistic use cases — not theoretical examples.

By the end of this course, students will be able to design and implement session-based features in a Next.js application with confidence, knowing when sessions are the right tool and — equally important — when they're not.

Course Philosophy

This course emphasizes:

- **Simplicity first:** Sessions are introduced as a natural extension of storage concepts students already know, not as a new system to fear.
 - **Use-case driven learning:** Every concept is tied to a real problem sessions solve — why teach what a session is before showing why anyone would need one?
 - **Healthy skepticism:** Students learn to validate and distrust session data by default, building habits that will serve them in more complex scenarios later.
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Course Statistics Summary

Section	Lessons	Estimated Words
00 - Welcome	1	~450
01 - What Is a Session?	3	~1,500
02 - Sessions and Storage	3	~1,600
03 - The Session Lifecycle	4	~2,100
04 - Sessions in Practice	3	~1,600
05 - Review and Assessment	2	~1,300

Section	Lessons	Estimated Words
06 - Wrap Up	1	~450
Total	18	~9,000

Section 00: Welcome

00.0 Welcome to Sessions 📖 (~450 words)

Learning Objectives:

- Understand what this course covers and why sessions matter in frontend development.
- Connect sessions to problems students have already encountered — apps that forget things they shouldn't.

Content Outline:

- The "forgetful app" problem: every page refresh wipes the slate clean — unless we do something about it.
- A quick preview of what sessions solve: remembering the user's state without asking them to log in.
- What students will build and know by the end of this course.
- A reminder of what's already been covered (LocalStorage/SessionStorage basics) and how this course builds on that.

Transitions: This lesson sets the stage. The next section dives straight into what a session actually is and why it can exist independently of any login process.

Key Principles:

- Sessions are a mechanism for giving an application short-term memory.
- Knowing when to use sessions — and when not to — is just as important as knowing how.

Examples:

- An e-commerce site that remembers your cart without you creating an account.
 - A multi-step form that preserves your progress if you accidentally navigate away.
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Section 01: What Is a Session?

01.0 Giving the App a Memory 📖 (~500 words)

Learning Objectives:

- Define what a session is in the context of a web application.
- Explain the problem sessions solve and why browsers don't do this automatically.
- Distinguish a session from a permanent record of user data.

Content Outline:

- The stateless nature of the web: HTTP doesn't remember who you are between requests.

- A session as a temporary "context" tied to a user's current visit.
- What makes a session a session: it starts, it persists across interactions during a visit, and it ends.
- Sessions vs. permanent storage: sessions are about the present visit, not a permanent record.

Transitions: Now that we know what a session is, the next lesson tackles a common misconception: that sessions are always tied to being logged in.

Key Principles:

- The web is stateless by design — sessions are one tool we use to layer state on top of that.
- A session represents a temporary window of context, not a permanent identity.

Examples:

- A news site that remembers which articles you've already read this visit — no account required.
 - A wizard-style form that tracks which step you're on between page navigations.
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01.1 Sessions Without Authentication 📖 (~500 words)

Learning Objectives:

- Explain why sessions do not require authentication to be useful.
- Identify real-world scenarios where anonymous sessions add value.
- Distinguish between a session (context) and authentication (identity).

Content Outline:

- The misconception: "session" doesn't mean "logged in."
- Anonymous sessions: tracking user context before any identity is known.
- Why this distinction matters — conflating sessions and auth leads to over-engineering simple problems.
- Examples of session data that has nothing to do with who the user is.

Transitions: With the concept clear, the next lesson moves to a hands-on question: given all that, what data actually belongs inside a session?

Key Principles:

- Authentication is about proving identity. Sessions are about remembering context. They are independent.
- Anonymous sessions are a legitimate and common pattern — not a shortcut or a workaround.

Examples:

- A language preference set on the first page load, stored in the session for the rest of the visit.
 - A shopping cart that accumulates items before the user ever creates an account.
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01.2 What Belongs in a Session? 📖 (~550 words)

Learning Objectives:

- Classify data as session-appropriate or not, using a clear decision framework.
- Apply the framework to realistic frontend scenarios.
- Avoid storing data in a session that should live elsewhere.

Content Outline:

- The key question: is this data temporary, visit-scoped, and non-sensitive?
- Data that belongs in a session: UI state, step progress, temporary preferences, anonymous context.
- Data that does NOT belong in a session: sensitive credentials, permanent user records, large payloads.
- Decision framework: temporary? visit-specific? safe if lost? → session. Otherwise → think again.

Transitions: Now that we know what to store in a session, the next section examines the tools available in the browser to actually do that storing.

Key Principles:

- Sessions are for temporary, visit-scoped context — not for data that needs to outlive the visit.
- Storing the wrong data in a session is a design problem, not just a technical one.

Examples:

- ☒ Step number in a multi-step checkout → session.
- ☒ Selected filter on a product listing page → session.
- ☒ User's email address → not a session (sensitive, should come from the server).
- ☒ All product data → not a session (too large, should be fetched from the API).

Hands-On Exercise: Students are given a list of 10 data items (e.g., "current onboarding step," "JWT token," "selected theme," "user password," "recently viewed product ID") and must classify each as session-appropriate or not, with a one-sentence justification.

Success Criteria:

- Correctly classifies at least 8 of 10 items.
 - Justifications reference the temporary/visit-scoped/non-sensitive framework.
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Section 02: Sessions and Storage

02.0 LocalStorage vs. SessionStorage (~550 words)

Learning Objectives:

- Distinguish LocalStorage from SessionStorage in terms of lifespan and scope.
- Explain which browser storage mechanism maps most naturally to session-like behavior.
- Choose the right storage tool for a given use case.

Content Outline:

- A quick recap of both APIs (students have seen these before — this is a focused comparison, not an introduction).
- The key difference: LocalStorage persists indefinitely; SessionStorage is tab-scoped and clears on close.
- When SessionStorage is the right choice: data you genuinely only need for this visit, in this tab.

- When LocalStorage is the right choice: preferences or state that should survive across visits.
- The nuance: not all "session data" has to live in SessionStorage — the right choice depends on how long the data should survive.

Transitions: Knowing the difference conceptually is one thing. The next lesson puts it into practice — actually reading and writing session data in code.

Key Principles:

- SessionStorage's tab-scoped, visit-limited nature makes it a natural fit for anonymous session data.
- The choice of storage mechanism should follow the expected lifespan of the data, not habit or convenience.

Examples:

- A multi-step form: step progress lives in SessionStorage (gone when the tab closes — intentional).
- A dark/light mode preference: lives in LocalStorage (should survive across visits).

02.1 Reading and Writing Session Data 📖 (~600 words)

Learning Objectives:

- Write and read data to/from SessionStorage using a consistent pattern.
- Handle JSON serialization and deserialization correctly.
- Build a reusable utility function for session data access in a Next.js project.

Content Outline:

- The basic API: `sessionStorage.setItem()`, `getItem()`, `removeItem()`.
- The JSON problem: storage only holds strings — objects must be serialized with `JSON.stringify()` and deserialized with `JSON.parse()`.
- Defensive reading: what happens when the key doesn't exist yet? Handle `null` gracefully.
- Building a small utility: `getSession(key)` and `setSession(key, value)` that wraps the boilerplate.

Transitions: Now that we know how to work with session storage correctly, the next lesson shows what happens when developers don't — specifically, when they store data they shouldn't.

Key Principles:

- Always serialize before writing, always parse after reading.
- Defensive defaults: assume the session might be empty and handle it gracefully.

Examples:

```
// Writing to session
sessionStorage.setItem('onboardingStep', JSON.stringify({ step: 2, completed:
['profile'] }));

// Reading from session (with safe fallback)
const raw = sessionStorage.getItem('onboardingStep');
const data = raw ? JSON.parse(raw) : { step: 1, completed: [] };
```

Hands-On Exercise: Students implement a `sessionStorage` utility module with three functions: `setSession(key, value)`, `getSession(key, defaultValue)`, and `clearSession(key)`. They then use it to persist and restore a simple object (e.g., a user's selected language and current wizard step) across a simulated page navigation.

Success Criteria:

- All three utility functions work correctly with object values.
 - `getSession` returns the provided default when the key is absent.
 - Data survives a `router.push()` navigation and is correctly restored on the destination page.
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02.2 Anti-Pattern: Storing Sensitive Data 📖 (~500 words)

Learning Objectives:

- Identify what makes data "sensitive" in the context of browser storage.
- Explain why browser storage is not a safe place for sensitive data.
- Describe the correct alternative for each class of sensitive data.

Content Outline:

- What counts as sensitive: credentials, tokens, personal identifiers, financial data.
- Why browser storage is not secure: it's readable by any JavaScript on the page — including third-party scripts.
- The XSS attack vector: a malicious script can quietly exfiltrate everything in `localStorage/sessionStorage`.
- The temptation: it's so easy to just store the token there. That's exactly why it's a trap.
- Correct alternatives: sensitive state should come from the server on each request, not be cached client-side in plain text.

Transitions: Now that we know what not to store and why, the next section shifts focus to the session lifecycle — how sessions start, persist, and end.

Key Principles:

- Convenience and security are often in tension. Browser storage is convenient — and therefore tempting for the wrong use cases.
- If data would cause harm if exposed, it does not belong in client-side storage.

Examples:

- ❌ Storing a user's raw password in `sessionStorage` "just temporarily."
 - ❌ Storing an API token in `localStorage` so it persists across visits.
 - ✅ Fetching the user's profile on each page load using a server-validated request.
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Section 03: The Session Lifecycle

03.0 Session Init 📖 (~500 words)

Learning Objectives:

- Describe what happens during the first interaction that creates a session.
- Identify the right moment to initialize a session in a Next.js application.
- Explain how to check whether a session already exists before creating one.

Content Outline:

- What "session init" means: the first time we write context to storage for this visit.
- When to init: on first meaningful user action (not on every render).
- The check-before-write pattern: always read first; only initialize if nothing is there.
- Using `useEffect` to check for an existing session when a component mounts.
- Starting a session with sensible defaults rather than empty state.

Transitions: With initialization covered, the next lesson looks at what happens in the middle of the session — how we keep reading and updating it as the user moves through the app.

Key Principles:

- Initialize once, update incrementally — don't recreate the session on every page visit.
- Always check before writing: treat existing session data as intentional, not accidental.

Examples:

```
useEffect(() => {  
  const existing = getSession('visitContext', null);  
  if (!existing) {  
    setSession('visitContext', { startedAt: Date.now(), pagesVisited: [] });  
  }  
}, []);
```

03.1 Keeping the Session Alive 📖 (~550 words)

Learning Objectives:

- Explain how session data persists across page navigations in a Next.js SPA.
- Update session data incrementally as the user interacts with the application.
- Avoid common mistakes that accidentally reset or corrupt session state.

Content Outline:

- How SPA navigation differs from full-page refreshes: `SessionStorage` survives `router.push()`.
- Reading the session on each relevant page/component mount and using it to restore state.
- Updating the session incrementally: write only the changed field, not the entire object every time.
- The merge pattern: read → update → write, never write from scratch.
- What causes accidental session resets: calling `removeItem` at the wrong moment, or overwriting instead of merging.

Transitions: Reading and updating are covered. The next lesson puts both into practice with a full working flow in Next.js.

Key Principles:

- SPA navigation does not clear SessionStorage — take advantage of that.
- Always merge updates into existing session data; never overwrite the whole object blindly.

Examples:

```
// Merge pattern – update one field without losing others
const current = getSession('visitContext', {});
setSession('visitContext', { ...current, lastPage: '/products' });
```

03.2 Building a Session Flow in Next.js 📖 (~650 words)

Learning Objectives:

- Implement a complete session init and read flow across two pages in Next.js.
- Use the session utility from Section 02 to manage state across navigations.
- Verify that session data persists correctly across `router.push()` calls.

Content Outline:

- Setting up a two-page scenario: a landing page (session init) and a dashboard page (session read).
- Initializing the session on the landing page with a timestamp and a default preference.
- Navigating to the dashboard and reading the session to display personalized content.
- Updating one session field on the dashboard without disturbing the rest.
- Testing the flow: open, navigate, verify data, navigate back, verify persistence.

Transitions: The flow is built and working. The next lesson covers the other end of the lifecycle — how and when to destroy a session.

Key Principles:

- Build flows around the check-before-write and merge patterns established in the previous lessons.
- Test session behavior by navigating, not just by reading the DevTools storage panel.

Examples:

```
// Landing page – init
useEffect(() => {
  const existing = getSession('appSession', null);
  if (!existing) {
    setSession('appSession', {
      startedAt: new Date().toISOString(),
      theme: 'light',
      lastPage: '/'
    });
  }
});
```



```
    }  
  }, []);  
  
  // Dashboard – read and update  
  useEffect(() => {  
    const session = getSession('appSession', {});  
    setTheme(session.theme ?? 'light');  
    setSession('appSession', { ...session, lastPage: '/dashboard' });  
  }, []);
```

Hands-On Exercise: Students build the two-page flow described above. The landing page initializes the session. The dashboard reads it and displays the `startedAt` timestamp and `theme` value. A button on the dashboard toggles the theme and writes the updated value back to the session. Navigating back to the landing page and then to the dashboard again should preserve the toggled theme.

Success Criteria:

- Session initializes only once (not on every visit to the landing page).
- Dashboard correctly reads and displays the `startedAt` and `theme` values.
- Theme toggle persists after navigating away and back.

03.3 Destroying a Session 🖥️ (~600 words)

Learning Objectives:

- Identify the correct moment to destroy a session.
- Clear session data cleanly using both targeted and full-clear approaches.
- Handle the UI state transition after session destruction.

Content Outline:

- When to destroy a session: explicit user action (e.g., "Start over"), end of a flow, or tab close (automatic for `SessionStorage`).
- `removeItem` vs. `clear`: targeted removal vs. wiping everything.
- Clearing session state and resetting UI in sync — they must happen together.
- The reset flow: clear session → reset local state → redirect or re-initialize.
- Edge case: what happens if the user tries to access a page that depends on session data after it's been cleared?

Transitions: The full lifecycle is now complete: init, update, destroy. The next section steps back to look at real-world use cases and a common trap developers fall into when working with sessions.

Key Principles:

- Destroying a session must be intentional and complete — half-cleared sessions lead to inconsistent UI.
- UI state and session state must be reset together; doing one without the other causes bugs.

Examples:

```
// Targeted removal
sessionStorage.removeItem('appSession');

// Full clear (use with caution)
sessionStorage.clear();

// Reset flow
const handleReset = () => {
  clearSession('appSession'); // utility from section 02
  setCurrentStep(1);
  router.push('/start');
};
```

Hands-On Exercise: Students add a "Reset session" button to the dashboard from the previous exercise. Clicking it clears the session, resets local state, and redirects to the landing page. On the landing page, the session should be re-initialized from scratch (as if it's a new visit). Students verify by checking that `startedAt` updates to the new timestamp.

Success Criteria:

- Session is fully cleared on reset (verified via DevTools Application tab).
- Redirect to landing page triggers a fresh session init.
- New `startedAt` timestamp is different from the previous one.

Section 04: Sessions in Practice

04.0 Real-World Use Cases 📖 (~500 words)

Learning Objectives:

- Identify common frontend scenarios where sessions are the right solution.
- Match each use case to the appropriate session pattern (init, update, destroy).
- Distinguish use cases where sessions are genuinely useful from those where they add unnecessary complexity.

Content Outline:

- Multi-step forms: preserving progress between steps without losing data on navigation.
- Onboarding flows: tracking which steps have been completed this visit.
- Anonymous preferences: language, currency, color scheme — before any account exists.
- Temporary search or filter state: remembering what the user was looking at when they navigated away.
- Use cases where sessions add no value: single-page interactions, data that should be permanent, data that must come from the server.

Transitions: Use cases identified — the next lesson builds one of them in code.

Key Principles:

- Sessions shine in multi-step, visit-scoped flows where losing state mid-process would frustrate the user.

- Not every UX problem needs a session. Default to the simplest solution.

Examples:

- A job application form that saves progress across the three-step flow, so navigating back doesn't lose answers.
 - A product catalog that remembers the active filters when the user clicks into a product and returns.
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04.1 Stateful UI with Session Data 📖 (~650 words)

Learning Objectives:

- Build a multi-step UI flow that uses session data to preserve progress.
- Read session data on mount to restore the user to their last position.
- Update the session on each step transition.

Content Outline:

- Scenario: a three-step onboarding flow (Step 1: name, Step 2: preferences, Step 3: summary).
- Initializing the session on Step 1 with default values.
- Writing the current step's data to the session before advancing.
- Reading the session on each step mount to pre-fill already-entered data.
- Showing a progress indicator driven by session data.

Transitions: The flow works — but what if the session data itself can't be trusted? The next lesson addresses that uncomfortable question.

Key Principles:

- Session-backed UI should always feel faster and smarter than stateless alternatives.
- Pre-filling data from the session is a UX gift — but only if the data was stored correctly.

Examples:

```
// Step 2 mount – read what was entered in Step 1
useEffect(() => {
  const session = getSession('onboarding', {});
  setName(session.name ?? '');
  setCurrentStep(session.currentStep ?? 1);
}, []);

// Advancing to Step 3 – write Step 2 data
const handleNext = () => {
  const session = getSession('onboarding', {});
  setSession('onboarding', { ...session, preferences: selectedPreferences,
    currentStep: 3 });
  router.push('/onboarding/step-3');
};
```

Hands-On Exercise: Students build a two-step form (Step 1: enter a name; Step 2: select a preference from three options). Session data is written at each step and read on mount. A "Back" button on Step 2 returns to Step 1 with the name field pre-filled from the session. Completing Step 2 logs the full session object to the console.

Success Criteria:

- Navigating back to Step 1 shows the previously entered name.
 - Completing Step 2 logs a session object containing both `name` and `preference`.
 - The session is not re-initialized if the user navigates back to Step 1.
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04.2 Anti-Pattern: Trusting Session Data Blindly 📖 (~500 words)

Learning Objectives:

- Explain why session data cannot be assumed to be valid, complete, or untampered.
- Apply defensive validation when reading session data before using it.
- Describe what can go wrong when session data is trusted without verification.

Content Outline:

- The temptation: "I wrote it, so it must be right."
- Why session data can be wrong: the user may have old data from a previous session, another script may have modified it, the schema may have changed after a deploy.
- The consequences of blind trust: rendering broken UI, using undefined values, silently skipping steps.
- Defensive reading: validate the shape and types of session data before using it.
- Sensible fallbacks: if the data doesn't look right, re-initialize rather than crash.

Transitions: With real-world patterns and their pitfalls covered, the final content section puts everything together in an assessment.

Key Principles:

- Treat session data the same way you'd treat any external input: verify before use.
- A defensive fallback is not a sign of weak code — it's a sign of experienced code.

Examples:

```
// Blind trust – dangerous
const session = getSession('onboarding', {});
setCurrentStep(session.currentStep); // could be undefined or an unexpected type

// Defensive read – correct
const session = getSession('onboarding', {});
const step = typeof session.currentStep === 'number' ? session.currentStep : 1;
setCurrentStep(step);
```

Section 05: Review and Assessment

05.0 Session Audit Challenge 📖 (~650 words)

Learning Objectives:

- Identify session-related bugs and anti-patterns in existing code.
- Apply fixes using the patterns learned throughout the course.
- Demonstrate understanding of the full session lifecycle in a practical review scenario.

Content Outline:

- Students are given a code snippet of a Next.js component that manages a session incorrectly.
- Problems present in the code: blind trust of session data, sensitive data in storage, no init check (overwrites on every mount), session never cleared.
- Task: identify each problem, explain why it's a problem, and rewrite the relevant section correctly.
- Review of the session utility patterns from Section 02 and the lifecycle patterns from Section 03.

Transitions: With the practical challenge complete, the final assessment checks conceptual understanding.

Key Principles:

- A session audit is a real professional skill — code reviews often catch exactly these kinds of issues.
- Every fix applied here maps to a pattern explicitly taught in this course.

Code Snippet for Review:

```
// Buggy component – students must find and fix all issues
useEffect(() => {
  // Problem 1: overwrites on every mount
  sessionStorage.setItem('user', JSON.stringify({
    email: userEmail, // Problem 2: sensitive data
    step: 1
  }));
}, []);

const advance = () => {
  const data = JSON.parse(sessionStorage.getItem('user'));
  // Problem 3: blind trust – no check if data is null or step is a number
  setStep(data.step + 1);
  // Problem 4: session is never cleaned up when flow completes
};
```

Hands-On Exercise: Students rewrite the component, fixing all four identified problems using the patterns from the course. Their corrected version must: check before init, exclude sensitive data, validate before use, and clear the session on flow completion.

Success Criteria:

- All four problems are identified and explained.
- The rewritten component uses the check-before-write, merge, defensive read, and clear-on-complete patterns.

- No sensitive data remains in `sessionStorage`.
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05.1 Session Knowledge Check 🧠 (~700 words)

Learning Objectives:

- Demonstrate understanding of session concepts, lifecycle, storage choices, and anti-patterns.
- Apply decision-making frameworks to new scenarios.
- Distinguish correct patterns from common mistakes.

Question Format: Multiple choice (5 questions)

Questions:

Question 1 A user fills in Step 1 of a 4-step checkout form, then navigates back using the browser's back button. When they return to Step 1, their data is gone. What is the most likely cause?

- A) The developer used `localStorage` instead of `sessionStorage`.
- B) The developer used React state without backing it up to session storage. ☒
- C) `sessionStorage` automatically clears when the user navigates back.
- D) The component re-mounted and overwrote the session data.

Question 2 Which of the following data items is the BEST candidate for session storage?

- A) The user's hashed password.
- B) The full product catalog (5,000 items).
- C) The active step in a multi-step onboarding wizard. ☒
- D) The user's account creation date.

Question 3 A developer writes the following code on every component mount:

`sessionStorage.setItem('flow', JSON.stringify({ step: 1 }));` What is wrong with this approach?

- A) `JSON.stringify` should not be used with `setItem`.
- B) The session is overwritten on every mount, losing any progress the user had made. ☒
- C) `setItem` does not accept string values.
- D) The key `'flow'` is reserved by the browser.

Question 4 After a user completes a multi-step flow, the session data is left in storage. What is the problem with this?

- A) The browser will throw an error if the session is not cleared.
- B) The next time the user starts the flow, they may see stale data from the previous run. ☒
- C) Session data automatically expires after 24 hours.
- D) There is no problem — leftover session data is harmless.

Question 5 A developer reads session data with this code: `const step = JSON.parse(sessionStorage.getItem('step'))`. The next line uses `step.current` directly. What defensive practice is missing?

- A) The developer should use `localStorage` instead.
- B) `JSON.parse` should be wrapped in a try/catch and the result validated before use. ☒

- C) Session data should never be parsed — it should be used as a string.
- D) The developer should call `sessionStorage.clear()` before reading.

Evaluation Criteria:

- 5/5: Strong grasp of session patterns, lifecycle, and anti-patterns.
 - 4/5: Solid understanding with one conceptual gap to revisit.
 - 3/5 or below: Review Sections 02 and 03 before moving on.
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Section 06: Wrap Up

06.0 Sessions Are Just the Beginning (~450 words)

Content Outline:

- Course recap: what students learned and built.
 - What a session is — and isn't.
 - The difference between `LocalStorage` and `SessionStorage`, and when to use each.
 - The full session lifecycle: init, update, and destroy.
 - Real-world use cases and the anti-patterns to avoid.
- Connection to the bigger picture: sessions are one layer in a broader state management story. Students now have a solid foundation for understanding more advanced patterns later in the program.
- What's coming next: interacting with external APIs — a natural extension of everything covered here, since API responses are often what sessions help cache or coordinate.
- Encouragement: managing state across navigations is one of the things that separates apps that feel polished from those that feel rough. Students now know how to build the polished kind.
- Resources for further exploration:
 - MDN Web Docs: Web Storage API
 - Next.js documentation: Client-side navigation and state
 - JavaScript.info: `LocalStorage` and `SessionStorage`

Note: This is conclusion only — no theory, exercises, or assessment.